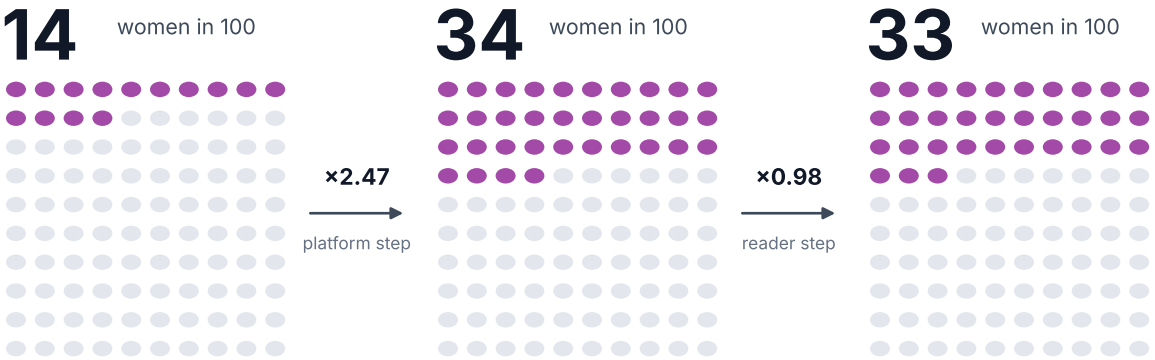
WHO GETS SEEN IN YOUR NEWS FEED?

Funnel Vision

An audit engine that follows every public figure named in the news from the story pool to the screen to the click. We benchmarked it on MIND-large, the field's standard dataset: 4,461 public figures in 29,309 stories, 97,592,931 recorded displays to 750,434 readers.

Out of every 100 people named in the stories, how many are women?



In the pool
Every story in circulation, counted once. What publishers supplied.

Shown
Every time the platform displayed a story to a reader. What the platform chose.

Clicked
Every time a reader clicked a story. What people chose.

Women became more visible, and more typecast.

We expected the platform to shrink women's share. It did the opposite: women were 13.6% of the people named in the story pool and 33.5% of the people readers were shown (95% interval ±0.03 points). But 98.5% of that rise came from the topic mix, led by celebrity and royal stories. 77% of women's appearances on screen were as entertainers or artists, up from 46% in the pool. Our typecast index (how far apart women's and men's roles are) rose from 0.52 to 0.61.

WHAT IS NEW IN THIS AUDIT

1 **People, not topics**
Feed audits usually measure topic diversity. We follow the 4,461 public figures inside the stories.

2 **The funnel ratio**
Share at one stage ÷ share at the stage before. It splits each gap into a platform step and a reader step.

3 **Mix or treatment**
A shift-share split separates a change in topic mix from different treatment within a topic.

4 **Nothing inferred**
No gender guessed from names. Only Wikidata records about public figures. Reader groups from clicks.

WHY THIS MATTERS IN CANADA

53%

of Canadians used social media for news in the past week, overtaking TV for the first time, even with news blocked on Facebook and Instagram.

Digital News Report 2026, Canada (Centre d'études sur les médias)

< 1/3

of the voices quoted in major Canadian news outlets belong to women and gender-diverse people.

Gender Gap Tracker, Informed Opinions and Simon Fraser University

99%

of the content on The Globe and Mail's digital pages is placed by its own AI system, Sophi, as reported by the vendor.

Mather Economics case study, The Globe and Mail and Sophi

The step rarely measured

Newsrooms get counted. Feeds rarely do. The EU's Digital Services Act already makes platforms disclose their recommenders' main parameters (Art. 27) and the largest offer a feed not based on profiling (Art. 38). We found no public dataset of what Canadians are shown, so we benchmark the engine on the field's standard logs first, which are US-centred.

Where this goes

A funnel scorecard for every algorithmic news feed in Canada. Each newsroom runs the engine on its own logs and publishes only the scores, so readers can see what the algorithm adds. Phase 2: the scorecard prototype, a within-section fix, and a pilot plan for a Canadian newsroom that already curates with an algorithm.

Data: Microsoft News Dataset (MIND-large, MSN News logs, November 9 to 15, 2019) and Wikidata. Charts and computed figures by the team, outside statistics cited where used. Sources, method and limitations: page 5.

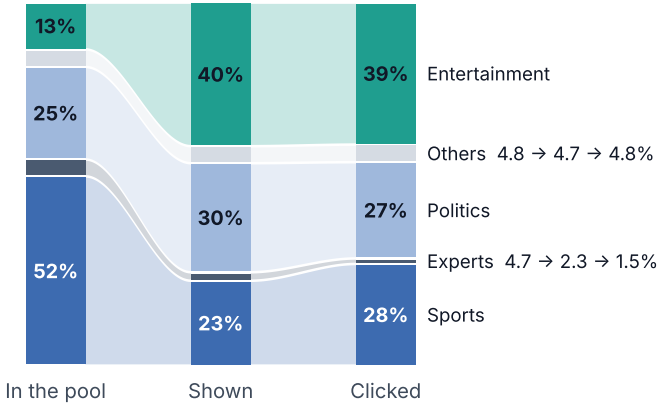
WHERE THE GAP OPENS · CHALLENGE QUESTIONS 2 AND 5

More women on screen, mostly through the topic mix

Women's share of people rose from 13.6% in the pool to 33.5% on screen, then held at 32.8% after clicks. 98.5% of the platform's change came from which subsections it showed more, led by celebrity and royal stories. Inside the same subsection it moved women's share by only +0.3 points.

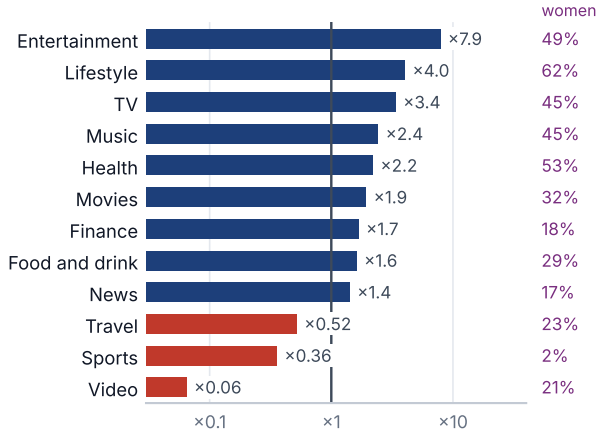
The platform reshapes who fills the screen

Share of all person appearances by role, one role per person from Wikidata occupations. Experts: science, health, academia, media and writing.



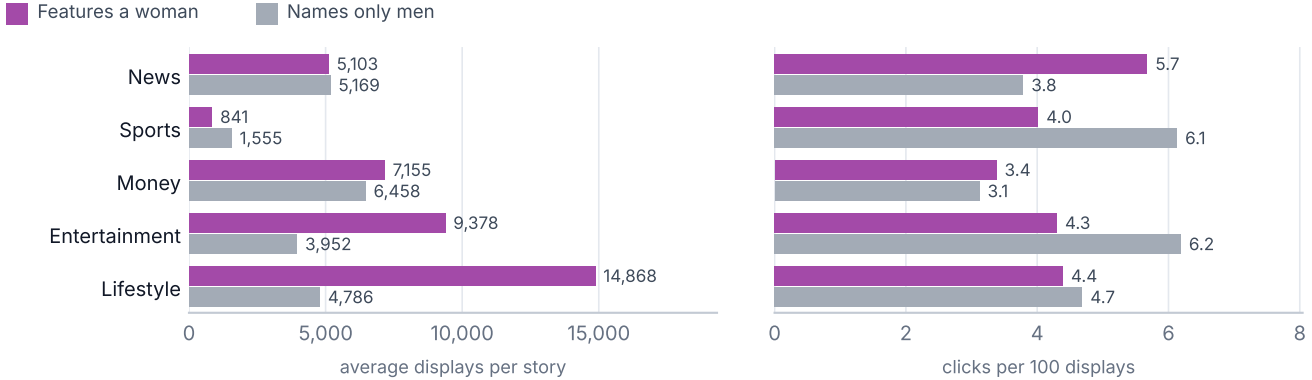
Which topics the platform over-shows

Funnel ratio of each topic's share of person appearances (shown ÷ pool). Purple: women's share of that topic's pool. Topic mix explains +19.7 of the +20.0 points women gained.



In the News section, readers click stories featuring women more than the platform shows them

Stories featuring a woman vs stories naming only men, by section. Left: average times each story was displayed. Right: clicks per 100 displays. Sections are defined on page 3.



News: about the same reach (5,103 vs 5,169 displays per story, medians 88.5 vs 89) but 50% more clicks per display. More reach can itself lower a click rate, so the Entertainment and Lifestyle gaps are weaker evidence.

The 20 most-shown people

10 of 20 are women. 9 of the 10 most-shown women are entertainers or royals. The other, Marie L. Yovanovitch, is in politics.

	share of all person appearances shown
Donald Trump	9.8%
Meghan, Duchess of Sussex	2.9%
Prince Harry, Duke of Sussex	1.7%
Joe Biden	1.3%
Jennifer Lopez	1.0%
Colin Kaepernick	1.0%
Catherine, Duchess of Cambridge	0.8%
Donald Trump Jr.	0.8%
Charles, Prince of Wales	0.8%
Kanye West	0.7%
Drake	0.6%
Chrissy Teigen	0.6%
Myles Garrett	0.6%
Taylor Swift	0.6%
John Legend	0.6%
Miranda Lambert	0.6%
Selena Gomez	0.6%
Sharon Osbourne	0.6%
Marie L. Yovanovitch	0.6%
Teresa Giudice	0.5%

9.8%

of all person appearances shown went to one man, Donald Trump.

Questions this raises

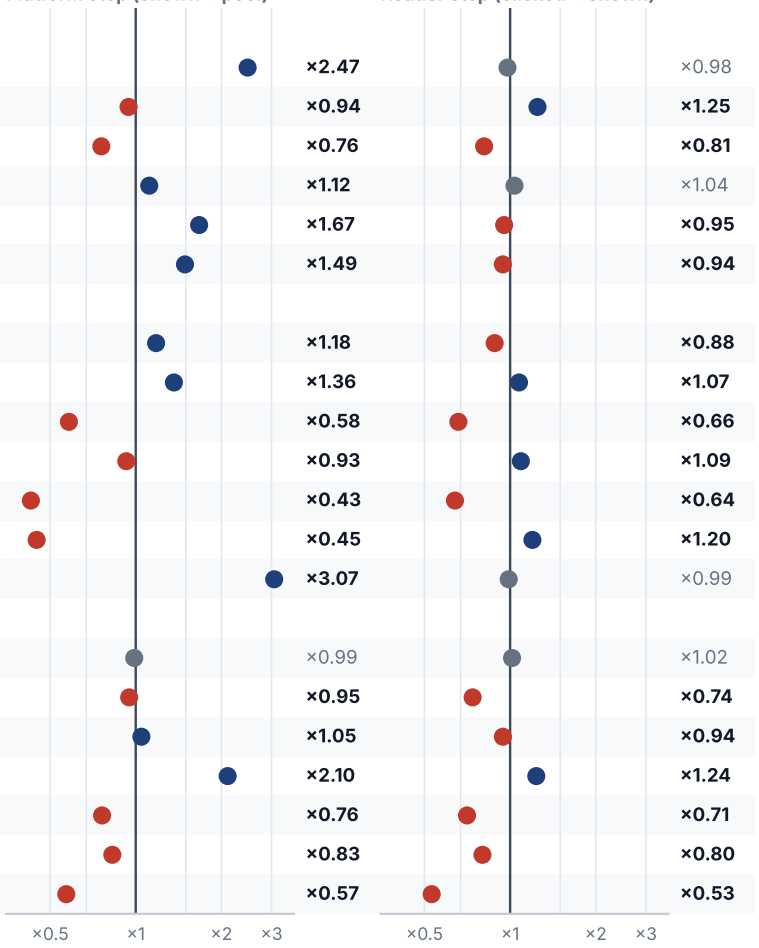
1. Why do News stories featuring women earn 50% more clicks per display, yet get no extra reach?
2. If a feed triples its entertainers and halves its experts, whose expertise do readers learn to trust?
3. Would a Canadian newsroom's feed show the same topic-mix effect, or a different one?

THE AUDIT SCORECARD · CHALLENGE QUESTIONS 2 AND 4

Who gains and who loses at each step

Entertainers gained the most at the platform step (×3.07). What readers miss: experts lost at both steps (science, health and academia ×0.58 then ×0.66, media and writing ×0.43 then ×0.64), and so did people from Africa, Latin America and Oceania. Canadian public figures lost mostly at the reader step (×0.74).

Group	In pool	Platform step (shown ÷ pool)	Reader step (clicked ÷ shown)
WOMEN, SHARE OF PEOPLE WITH A RECORDED GENDER			
Women (all sections)	13.6%	**×2.47**	×0.98
Women in News	17.3%	**×0.94**	**×1.25**
Women in Sports	2.2%	**×0.76**	**×0.81**
Women in Money	17.9%	**×1.12**	×1.04
Women in Entertainment	32.6%	**×1.67**	**×0.95**
Women in Lifestyle	44.1%	**×1.49**	**×0.94**
ROLES, SHARE OF ALL PEOPLE NAMED			
Politics & government	25.5%	**×1.18**	**×0.88**
Business	1.3%	**×1.36**	**×1.07**
Science, health & academia	2.3%	**×0.58**	**×0.66**
Law & justice	1.2%	**×0.93**	**×1.09**
Media & writing	2.4%	**×0.43**	**×0.64**
Sports	52.1%	**×0.45**	**×1.20**
Entertainment & arts	12.9%	**×3.07**	×0.99
CITIZENSHIP, SHARE OF PEOPLE WITH ONE RECORDED			
United States	82.6%	×0.99	×1.02
Canada	2.8%	**×0.95**	**×0.74**
Europe	9.6%	**×1.05**	**×0.94**
Asia	1.4%	**×2.10**	**×1.24**
Africa	0.7%	**×0.76**	**×0.71**
Latin America & Caribbean	2.3%	**×0.83**	**×0.80**
Oceania	0.7%	**×0.57**	**×0.53**



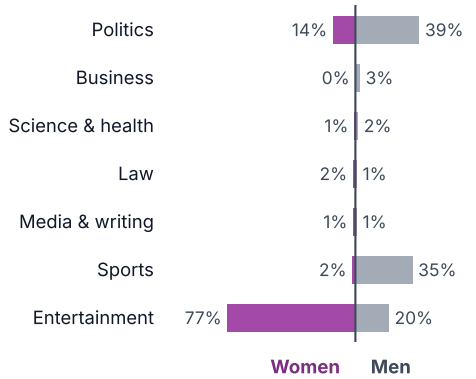
gained · lost · within 5% of ×1

Funnel ratio: a group's share at a stage ÷ its share at the stage before. ×1 means the step kept the group as it found it. Every 95% interval excludes ×1, but intervals resample readers, not stories, so they are too narrow for groups seen in few stories. Gray marks shifts too small to matter.

Sections group MIND's categories: Money is finance, Entertainment is entertainment, TV, music, movies, video and kids, Lifestyle is lifestyle, food, travel, health, autos and weather. MSN's video channel is 48% of Entertainment's pool appearances but 2% of what it shows, so the Women in Entertainment ratio largely reflects that mix.

As whom do women appear on screen?

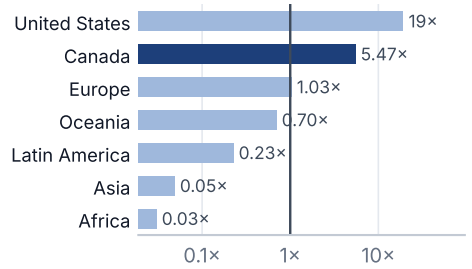
Share of women's and of men's appearances shown, one role per person from Wikidata occupations. Entertainers: 46% of women's appearances in the pool, 77% on screen.



Typecast index (0 = same roles, 1 = no overlap): 0.52 in the pool, 0.61 shown, 0.61 clicked.

Whose world?

Citizenship of the people named, not where stories happen, grouped by Wikidata continent (dual citizens split evenly). Share of people shown ÷ share of the 2019 world population (UN). Asia, Africa and Latin America: 85% of the world, 5.3% of the people shown.



Not measured: race, ethnicity and Indigenous identity. Wikidata records ethnicity for under 1.2% of people (Kronk 2026), and guessing would break the no-inference rule.

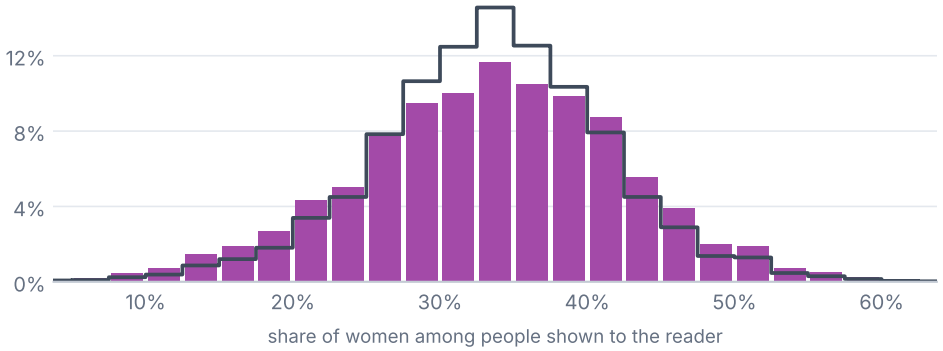
YOUR FEED, YOUR GAP · CHALLENGE QUESTIONS 1 AND 3

Your past clicks predict who you see

Readers see different worlds. Among readers shown at least 20 named people, one in ten saw 21.3% women or fewer and one in ten 44.7% or more. Sports-first readers saw 28.2% women, entertainment-first readers 38.0%. Past clicks predict it (rank correlation 0.37, against 0.02 with personalization shuffled away), and twice as many readers see under 10% women as when it is shuffled away (0.86% against 0.44%).

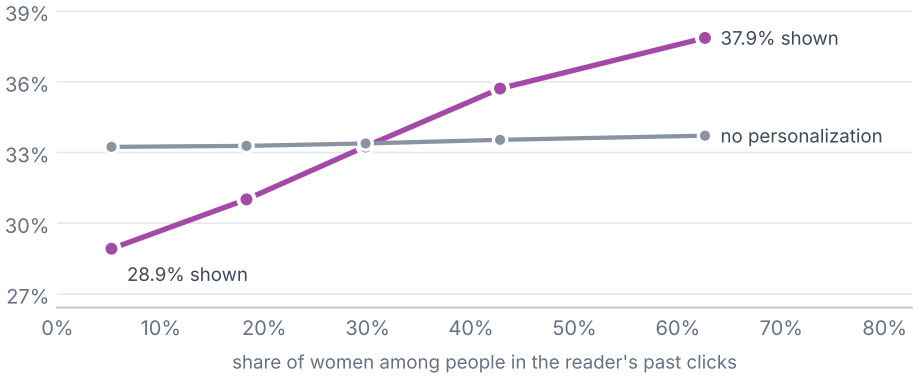
How often each reader saw women in the news

Share of women among people shown to each of 525,715 readers (purple) vs the same feeds shuffled between readers active in the same hour, which removes personalization but keeps the news cycle (gray line, 30 shuffles). Spread: ×1.17 wider than chance. In 14.7% of feed refreshes that named people, none was a woman.



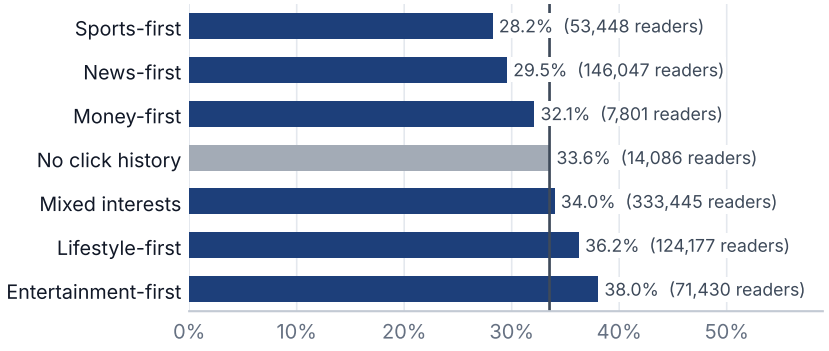
Past clicks predict what you are shown

337,501 readers grouped into fifths by the share of women in their past clicks. Purple: what they were shown. Gray: the same readers with personalization shuffled away.



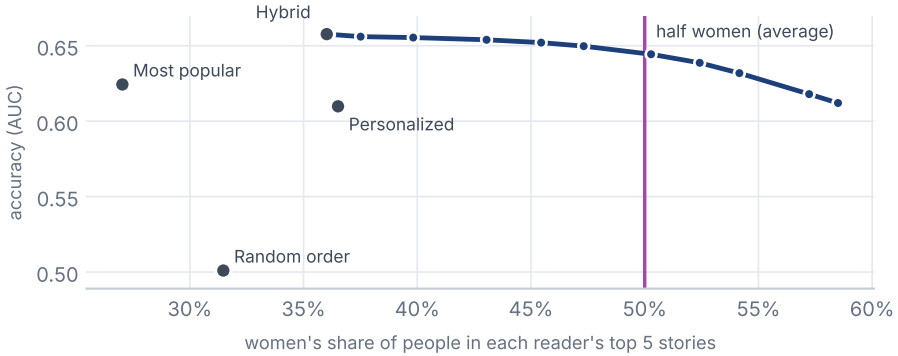
Reader communities, defined only by what they click

MIND has no reader demographics, so we infer none. A community is the section behind at least half of a reader's past clicks. Gray: readers with no click history in MIND, a natural baseline. Line: all readers, 33.5%. Light and heavy readers saw 33.2% and 33.6%, so the gap follows interests, not activity.



Does relevance have to cost representation?

Baselines trained on the first six days, tested on 100,000 feed refreshes from day seven. Line: our best model (hybrid) with a growing boost for stories featuring a woman. Top-5 women: personalized 36.5%, random 31.5%, most popular 27.0% (lowest).



Raising women to half the people in top-5 lists, on average, costs 1.3 AUC points (0.658 to 0.644) and cuts lists that name people but no woman from 23% to 9%. 59% of stories featuring a woman are Entertainment or Lifestyle, so a blanket boost leans toward them. Phase 2 tests a boost within each section.

Data: Microsoft News Dataset (MIND-large, MSN News logs, November 9 to 15, 2019) and Wikidata. Charts and computed figures by the team, outside statistics cited where used. Sources, method and limitations: page 5.

DATA SOURCES, METHODOLOGY AND LIMITATIONS · CHALLENGE QUESTION 5

How we measured it, and what the data cannot tell us

Every computed number in this report can be reproduced from public data with our two Python scripts. Any platform that logs stories, the people they name, and what was shown and clicked can compute the same six indicators.

MIND-large
750,434 readers, 2,609,219 feed refreshes, 97,592,931 displays

→

Entity links
Microsoft's Wikidata links in every headline and abstract

→

Wikidata
10,387 linked people in the catalog, 4,461 in stories shown

→

Three stages
Pool, shown, clicked. Funnel ratios with reader bootstrap

→

Tests
Topic-mix split, shuffle null, feedback loop, recommender trade-off

1 Data sources

Microsoft News Dataset (MIND-large, Wu et al., ACL 2020, research license), the benchmark behind 57% of news recommender papers from 2022 to 2024: anonymized MSN News logs of 750,434 readers, November 9 to 15, 2019, training and validation splits (the test split has no click labels). Wikidata (CC0, retrieved 2026-10) for each linked person's gender (P21), occupation (P106), citizenship (P27) and that country's continent (P30). UN World Population Prospects 2019.

2 What it represents, and what it does not

What Microsoft News displayed and what its readers clicked during one week of 2019, in English. Reader location is not recorded and 82% of the people shown are US citizens, so this is a US-centred sample, not a Canadian one. It sees only public figures with a Wikidata page that Microsoft's tagger linked: 29% of stories name at least one. A mention is not a quote. MSN's displays may mix editors' and algorithms' choices. MIND has no publisher or stance labels, so we measure whose voices appear (roles), not sources or viewpoints.

3 Our proxy for information exposure

A display: one story shown to a reader inside a feed refresh (MIND's impression log). We assume a displayed story was seen, since MIND has no scroll or position data. Each person Microsoft's entity links find in the headline or abstract is one person appearance. The pool counts each story in circulation once. Clicked counts the displays a reader clicked.

4 Limitations and biases

In the data: MIND samples MSN readers with at least 5 clicks in six weeks, so light readers are under-represented. People named only in a story's body are missed. Wikidata under-records women (28.7% of humans with a stated gender, Kronk 2026), so we read changes between stages, not levels, and its attributes are as of 2026 (99% have an occupation, 97% a citizenship). Gender is as recorded, never guessed from names, with 11 people outside the woman and man categories. Readers have no demographics, so none are inferred. In our method: intervals resample readers, not stories, so they are too narrow for groups seen in few stories. One role per person by a fixed rule (Prince Harry lands in Sports). Results are associations, not causes.

5 Why this metric is appropriate

Fairness needs a reference point. Our main one is the pool: the funnel ratio compares each stage with the one before it, so it isolates which step changes representation and judges the platform only on what it adds. We also compare with world population (visibility index) and with parity (half women, the trade-off target). Six indicators of bias or unequal outcomes: funnel ratio per group (women ×2.47 at the platform step), typecast index (Duncan dissimilarity over roles, 0.52 to 0.61), visibility index (Africa 0.03×), top-10 share of attention (20.9%), narrow-feed rate (0.86% of readers under 10% women, 0.44% by chance) and feed refreshes that name people but no woman (14.7%). Shifts within 5% of ×1 count as no change. 300 reader bootstraps (the pool is a full count, so intervals cover the shown and clicked stages) and 30 same-hour shuffles.

How this improves on existing work

Gender Gap Tracker	who Canadian outlets quote at publication, with gender assigned from names.
RADio	recommendations vs the pool or reader history: topics, viewpoints, minority voices.
Vrijenhoek (2023)	MIND's pipeline stages, for article categories only.
FairRec, EB-NeRD	what men and women readers receive, which needs reader demographics.
Funnel Vision	recorded attributes of the people in stories, pool to click, no reader demographics.

Robustness: does the result survive other choices?

	Pool	Shown	Clicked	Platform	Reader
Main analysis	13.6%	33.5%	32.8%	**×2.47**	×0.98
Excluding Sports	26.5%	41.0%	41.6%	**×1.55**	×1.01
Pool = full news catalog	12.0%	33.5%	32.8%	**×2.80**	×0.98
Headlines only	12.5%	31.4%	30.5%	**×2.51**	×0.97
Without the 10 most-shown people	14.3%	36.4%	35.6%	**×2.55**	×0.98
Story level, any woman	17.8%	39.0%	36.8%	**×2.19**	×0.94
First six days only	13.6%	33.5%	33.6%	**×2.46**	×1.00
Last day only	20.3%	33.9%	28.2%	**×1.67**	×0.83

Every check keeps the platform step above ×1.5. All entity links have confidence of at least 0.9, so a confidence filter changes nothing.

Is the data sufficient?

For the headline, yes: 51,418,192 person appearances shown, and all 7 alternative checks agree on direction. Coverage is the limit: 29% of stories and 33% of displays name a linked person (4,461 people in stories shown, 10,387 in the catalog, 1,725 of them women). For groups under 1% of the pool (Africa and Oceania), read the ratios as indicative.

Team and roles

Soufiane Amribt	Computer Science (cybersecurity), UNB. Data pipeline, statistics and recommender experiment
Jenny Kim	Software Engineering, UNB. Independent reproduction of the results on a second machine, code review
Josh Bacon	Software Engineering, UNB. Fact-check of findings against the data, review of limitations and wording

Sources

Wu et al. 2020, MIND, ACL. Vrijenhoek et al. 2024, RADio*, ACM TORS. Vrijenhoek 2023, Do you MIND?, BIAS. FairRec 2021, arXiv:2006.16742. EB-NeRD 2024, arXiv:2410.03432. Asr, Taboada et al. 2021, Gender Gap Tracker, PLOS ONE. CEM 2026, Digital News Report Canada. Mather Economics, Globe and Mail and Sophi. Kronk 2026, arXiv:2609.22375. Diversity-aware news recommender datasets 2025, arXiv:2510.05952. Duncan and Duncan 1955, American Sociological Review. EU Digital Services Act, Regulation (EU) 2022/2065, Art. 27 and 38. UN World Population Prospects 2019.